NorthRoss_MetaAnalysis

Clean up data

```r
library(metafor)
```

```
## Loading required package: Matrix

## Loading required package: metadat

## Loading required package: numDeriv

##
## Loading the 'metafor' package (version 5.0-1). For an
## introduction to the package please type: help(metafor)
```

```r
library(orchaRd)
```

```
##
## Loading the 'orchaRd' package (version 2.0). For an
## introduction and vignette to the package please see: https://daniel1noble.github.io/orchaRd/
```

```r
df <- read.csv("ExtractedData_2026-05-31.csv")
# Convert all variances to SD
table(df$Biodiversity_Variance_unit)
```

```
##
##    DS SD SE
##  7  1 24 18
```

```r
df$Control_SD = df$Control_var
df$Treatment_SD = df$Treatment_var
for(i in 1:nrow(df)){
  dfi <- df[i,]
  if(dfi$Biodiversity_Variance_unit == "SE"){

    dfi$Control_SD = dfi$Control_var * sqrt(dfi$Control_n)
    dfi$Treatment_SD = dfi$Treatment_var * sqrt(dfi$Treatment_n)
  }
  df[i,] <- dfi
}

# Note -some of the variances are too high and need to be corrected
```

```r
# Combine peatland types
df$Peatland_type <- replace(df$Peatland_type,
                            grepl(" fen", df$Peatland_type),
                            "Fen")
table(df$Peatland_type)
```

```
##
##                      Bog          Fen Spruce swamp
##            2          22           21            5
```

```r
# Combine functional groups
df$FG_Group <- replace(df$FG_Group,
                       grepl(" moss", df$FG_Group),
                       "Moss")

# Convert some variables to factors
df$Peatland_type <- factor(df$Peatland_type)
df$FG_Group <- factor(df$FG_Group)

# Convert to numeric
df$WTD_Difference <- as.numeric(df$WTD_Difference)
```

Func Groups as mod

```r
# Let's try doing all functional group data then separating them as a variable
fg <- subset(df, df$SR_or_FG=="FG")

# Remove empty
fg <- subset(fg, fg$FG_Group != "")
# Remove "vascular plant" (not really useful for comparison)
fg <- subset(fg, fg$FG_Group != "Vascular plant")

fg$FG_Group <- factor(fg$FG_Group)
# Get effect size
fgES <- escalc(measure = "SMDH", # I think this is best,
               # since many of the studies use different units and
               # measurement techniques and different sample sizes
               m2i = Control_mean,
               sd2i = Control_SD,
               n2i = Control_n,
               m1i = Treatment_mean,
               sd1i = Treatment_SD,
               n1i = Treatment_n,
               data = fg)
# Try a one-variable with FG
fg1 <-rma.mv(
  yi=yi,
  V=vi,
  mods=~FG_Group -1,
  random= ~1|PaperID/ID,
```

```r
  method="REML",
  digits=4,
  data=fgES)
```

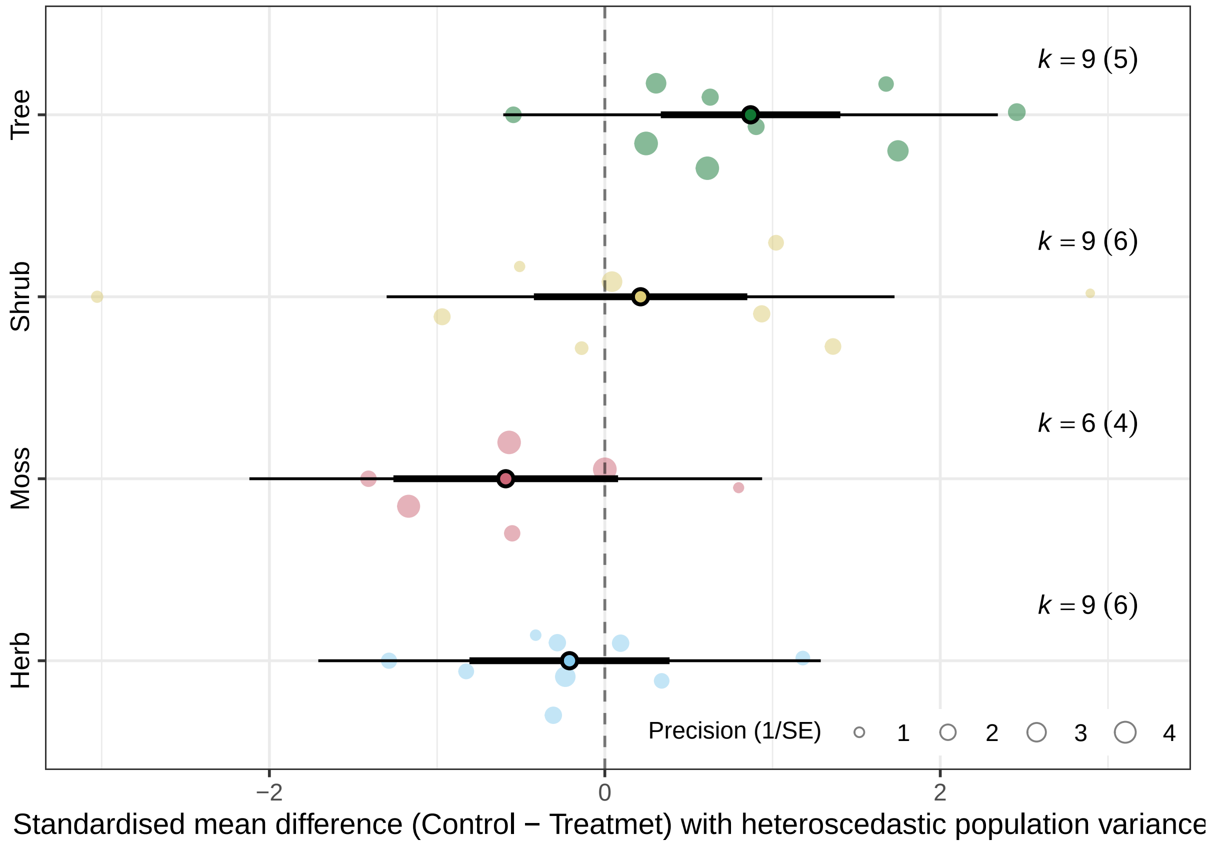
```
## Warning: 1 row with NAs omitted from model fitting.
```

```r
summary(fg1)
```

```
##
## Multivariate Meta-Analysis Model (k = 33; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
## -40.9583   81.9166   93.9166  102.1203   97.7348
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed      factor
## sigma^2.1  0.0000  0.0001      9     no     PaperID
## sigma^2.2  0.4914  0.7010     33     no  PaperID/ID
##
## Test for Residual Heterogeneity:
## QE(df = 29) = 95.4596, p-val < .0001
##
## Test of Moderators (coefficients 1:4):
## QM(df = 4) = 14.0232, p-val = 0.0072
##
## Model Results:
##
##                 estimate      se     zval    pval    ci.lb   ci.ub
## FG_GroupHerb     -0.2106  0.3044  -0.6919  0.4890  -0.8072  0.3860
## FG_GroupMoss     -0.5908  0.3417  -1.7290  0.0838  -1.2606  0.0789   .
## FG_GroupShrub     0.2132  0.3246   0.6568  0.5113  -0.4230  0.8494
## FG_GroupTree      0.8688  0.2731   3.1818  0.0015   0.3336  1.4040  **
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# only significant change is in trees at p=0.01

orchard_plot(fg1,
             mod = "FG_Group",
             group = "PaperID",
             xlab = "Standardised mean difference (Control - Treatmet) with heteroscedastic population v
```



Highly significant increase in trees, decrease in moss at p=0.1. Non-significant decrease in herbs (unexpected) and increase in shrubs (expected).

Try using CaseID as a random effect and group in plot:

```
# Try a one-variable with FG
fg3 <-rma.mv(
  yi=yi,
  V=vi,
  mods=~FG_Group -1,
  random= ~1|PaperID/Case_ID/ID,
  method="REML",
  digits=4,
  data=fgES)
```

```
## Warning: 1 row with NAs omitted from model fitting.
```

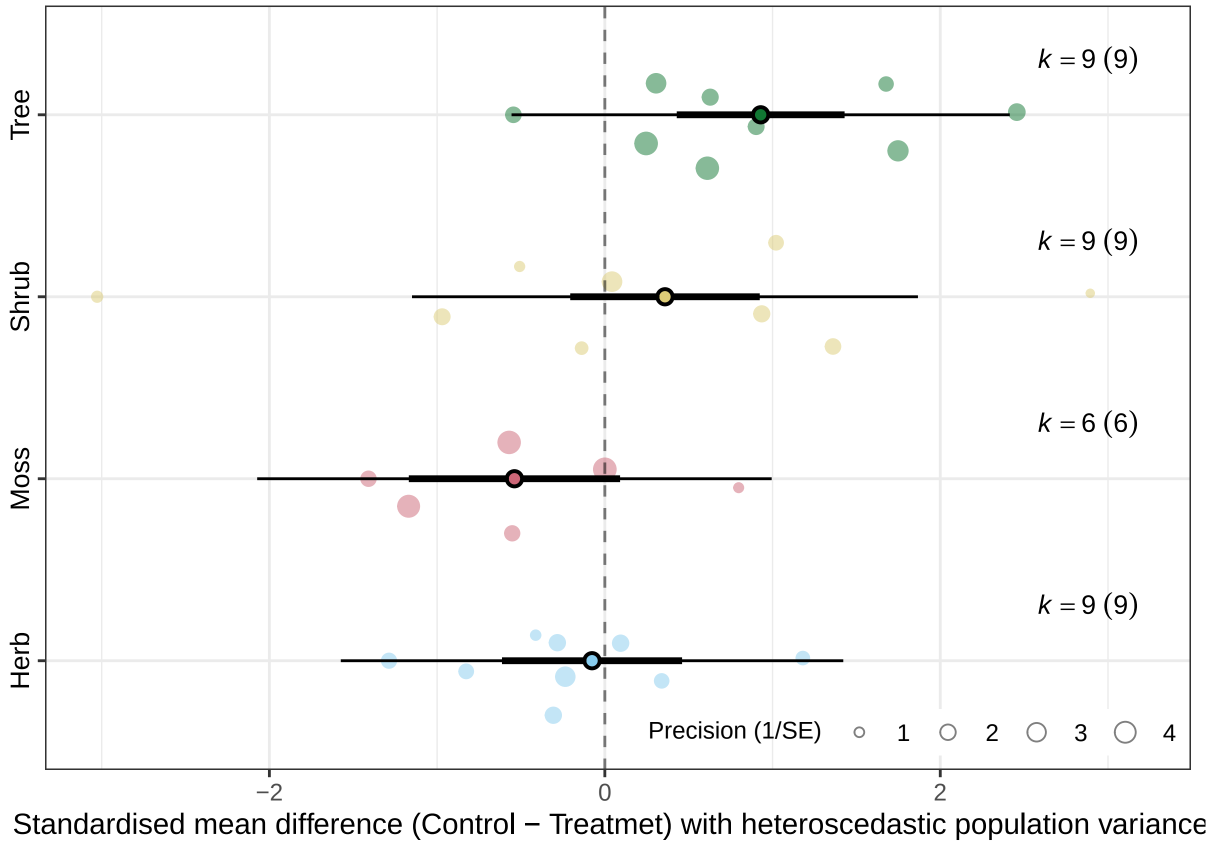
```
summary(fg3)
```

```
##
## Multivariate Meta-Analysis Model (k = 33; method: REML)
##
##    logLik  Deviance       AIC       BIC      AICc
## -37.8030   75.6061   89.6061   99.1771   94.9394
##
## Variance Components:
```

```
##
##             estim    sqrt  nlvls  fixed             factor
## sigma^2.1  0.0000  0.0000      9     no            PaperID
## sigma^2.2  0.4804  0.6931     16     no     PaperID/Case_ID
## sigma^2.3  0.0287  0.1695     33     no  PaperID/Case_ID/ID
##
## Test for Residual Heterogeneity:
## QE(df = 29) = 95.4596, p-val < .0001
##
## Test of Moderators (coefficients 1:4):
## QM(df = 4) = 21.9710, p-val = 0.0002
##
## Model Results:
##
##                estimate      se     zval    pval    ci.lb   ci.ub
## FG_GroupHerb    -0.0763  0.2740  -0.2787  0.7805  -0.6133  0.4606
## FG_GroupMoss    -0.5390  0.3214  -1.6771  0.0935  -1.1689  0.0909    .
## FG_GroupShrub    0.3586  0.2883   1.2438  0.2136  -0.2065  0.9237
## FG_GroupTree     0.9291  0.2553   3.6392  0.0003   0.4287  1.4295  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# only significant change is in trees at p=0.01

orchard_plot(fg3,
             mod = "FG_Group",
             group = "Case_ID",
             xlab = "Standardised mean difference (Control - Treatmet) with heteroscedastic population v
```




```r
# Try with WTD difference as another mod
fg2 <-rma.mv(
  yi=yi,
  V=vi,
  mods = ~ FG_Group * WTD_Difference -1,
  random= ~1|PaperID/Case_ID/ID,
  method="REML",
  digits=4,
  data=fgES)
```

```
## Warning: 1 row with NAs omitted from model fitting.
```

```r
summary(fg2)
```

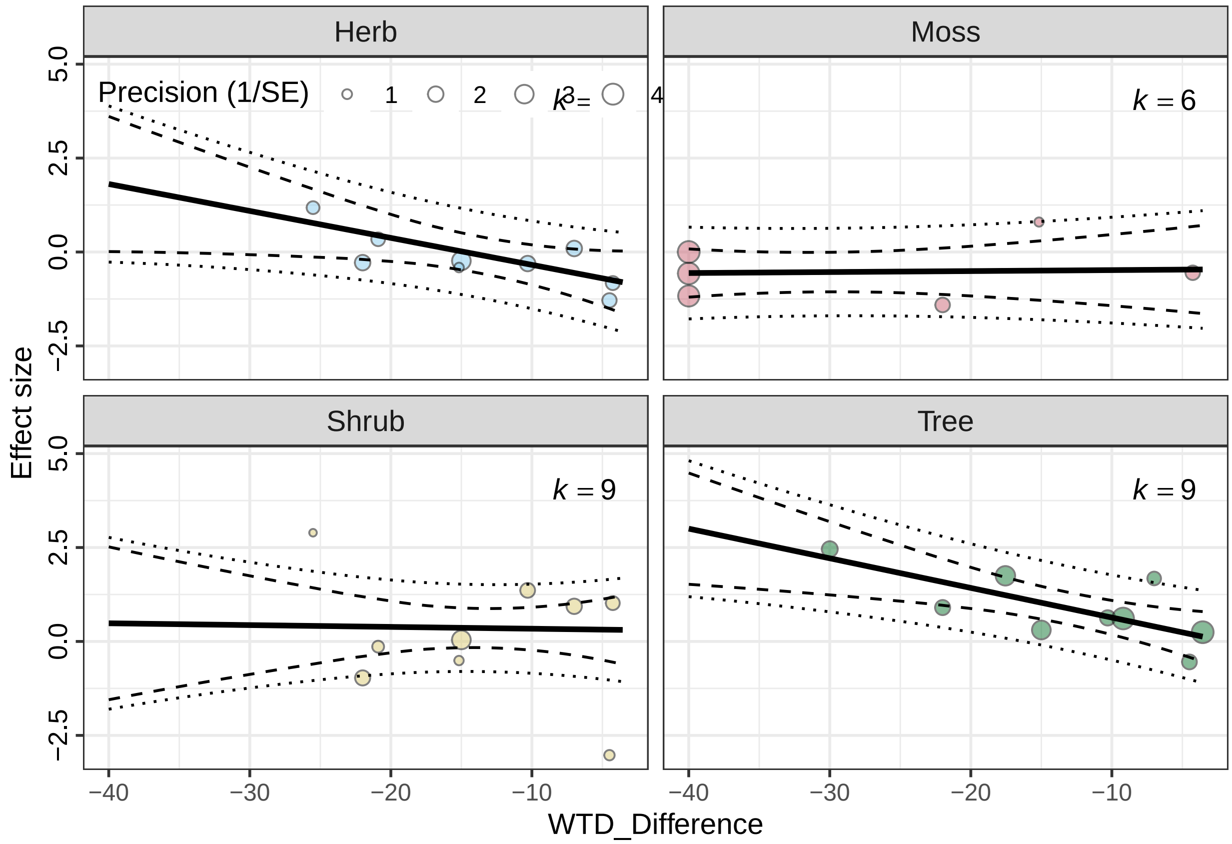
```
##
## Multivariate Meta-Analysis Model (k = 33; method: REML)
##
##    logLik  Deviance        AIC        BIC       AICc
## -30.1514   60.3027   82.3027   95.7104   102.6104
##
## Variance Components:
##
##               estim    sqrt  nlvls  fixed             factor
## sigma^2.1    0.0000  0.0000      9     no            PaperID
## sigma^2.2    0.2814  0.5305     16     no    PaperID/Case_ID
```

```
## sigma^2.3  0.0000  0.0000     33     no  PaperID/Case_ID/ID
##
## Test for Residual Heterogeneity:
## QE(df = 25) = 67.4360, p-val < .0001
##
## Test of Moderators (coefficients 1:8):
## QM(df = 8) = 40.2847, p-val < .0001
##
## Model Results:
##
##                               estimate      se     zval    pval    ci.lb
## FG_GroupHerb                   -1.0590  0.5188  -2.0410  0.0413  -2.0759
## FG_GroupMoss                   -0.4559  0.6592  -0.6916  0.4892  -1.7479
## FG_GroupShrub                   0.2911  0.5767   0.5047  0.6138  -0.8393
## FG_GroupTree                   -0.1524  0.4163  -0.3661  0.7143  -0.9684
## WTD_Difference                 -0.0717  0.0335  -2.1404  0.0323  -0.1374
## FG_GroupMoss:WTD_Difference     0.0743  0.0373   1.9946  0.0461   0.0013
## FG_GroupShrub:WTD_Difference    0.0669  0.0420   1.5935  0.1111  -0.0154
## FG_GroupTree:WTD_Difference    -0.0071  0.0390  -0.1834  0.8545  -0.0835
##                                 ci.ub
## FG_GroupHerb                  -0.0420  *
## FG_GroupMoss                   0.8362
## FG_GroupShrub                  1.4214
## FG_GroupTree                   0.6635
## WTD_Difference                -0.0060  *
## FG_GroupMoss:WTD_Difference    0.1473  *
## FG_GroupShrub:WTD_Difference   0.1492
## FG_GroupTree:WTD_Difference    0.0692
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fg2_plot <- mod_results(fg2, mod = "WTD_Difference", group = "PaperID", by = "FG_Group")

bubble_plot(fg2_plot, group = "PaperID",  mod = "WTD_Difference", xlab = "WTD_Difference")
```

```
## Ignoring unknown labels:
## * parse : "TRUE"
```



Herb production was significantly negatively affected by WTD difference at the p=0.01 level - studies with small changes in WTD resulted in slight decreases in herb abundance,while high WTD showed increases. This makes sense - studies that didn't really change WTD much didn't have a strong effect on shrubs, but large decreases in WTD had increased shrub production. (note that "WTD_Difference" above is actually FG_GroupHerb:WTD_Difference)

Mosses also had a significant increase at the p=0.01 level, but it's a very small increase (0.0765 units/cm), doesn't really tell us much. Herbs were also significant

Looking at just herbs with peatland type as mod

```
# Filter to just herbs
# Later I will compare the effect on herbs, trees and mosses
table(df$FG_Group)
```

```
##
##                              Herb          Moss          Shrub          Tree
##              3                  9             6              9            10
## Vascular plant
##              3
```

```
herbs <- subset(df,df$FG_Group=="Herb")
```

```r
# Get effect size
datES <- escalc(measure = "SMDH", # I think this is best,
                    # since many of the studies use different units and
                    # measurement techniques and different sample sizes
                m2i = Control_mean,
                sd2i = Control_SD,
                n2i = Control_n,
                m1i = Treatment_mean,
                sd1i = Treatment_SD,
                n1i = Treatment_n,
                data = herbs)

# Grand mean
grandmean1 <-rma.mv(
  yi=yi,
  V=vi,
  random= ~1|PaperID/ID,
  method="REML",
  digits=4,
  data=datES)
summary(grandmean1)
```

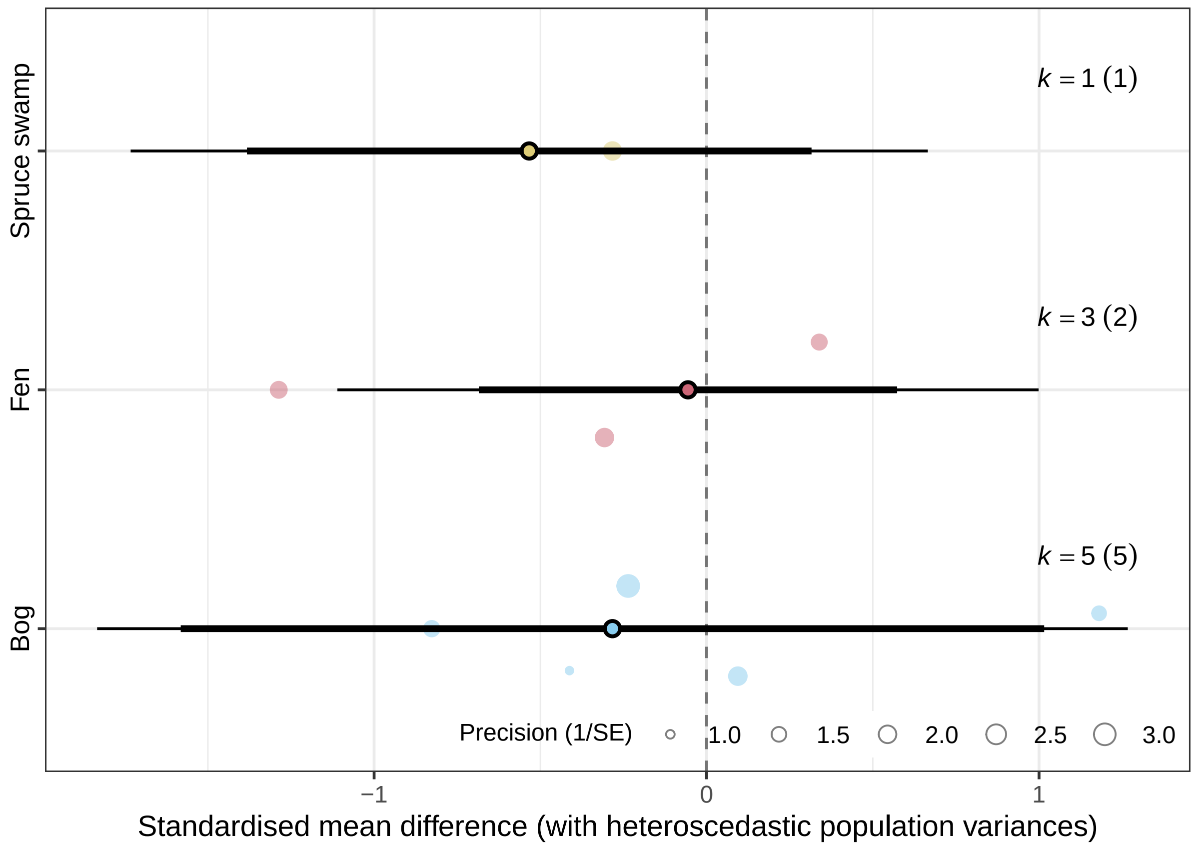
```
##
## Multivariate Meta-Analysis Model (k = 9; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##  -7.6510   15.3020   21.3020   21.5403   27.3020
##
## Variance Components:
##
##              estim    sqrt  nlvls  fixed      factor
## sigma^2.1  0.0000  0.0000      6     no     PaperID
## sigma^2.2  0.0000  0.0000      9     no  PaperID/ID
##
## Test for Heterogeneity:
## Q(df = 8) = 9.2191, p-val = 0.3242
##
## Model Results:
##
## estimate      se     zval    pval    ci.lb   ci.ub
##  -0.2290  0.1770  -1.2938  0.1957  -0.5759  0.1179
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Try a one-variable with Peatland type
peat1 <-rma.mv(
  yi=yi,
  V=vi,
  mods=~Peatland_type,
  random= ~1|PaperID/ID,
  method="REML",
  digits=4,
```

```
  data=datES)
```

```
## Warning: Redundant predictors dropped from the model.
```

```
orchard_plot(peat1,
             mod = "Peatland_type",
             group = "PaperID",
             xlab = "Standardised mean difference (with heteroscedastic population variances)")
```



The grand mean of the effect size was -0.2296, but only had a p-value of 0.1946, indicating that there was no significant effect of WLD on herb production.

Species Richness

We didn't get a lot of data on species richness, but let's take a look at it:

```
sr <- subset(df, df$SR_or_FG=="SR")
# Remove NA
sr <- subset(sr, !is.na(sr$Control_mean))

print(length(sr))
```

```
## [1] 40
```

```r
# only 9 records...

# factorize peatland type
sr$Peatland_type <- factor(sr$Peatland_type)

# Get effect size
srES <- escalc(measure = "ROM", # Since we are looking at positive numbers similar in scale to each oth
                m2i = Control_mean,
                sd2i = Control_SD,
                n2i = Control_n,
                m1i = Treatment_mean,
                sd1i = Treatment_SD,
                n1i = Treatment_n,
                data = sr)
# grand mean
sr1 <-rma.mv(
  yi=yi,
  V=vi,
  random= ~1|PaperID/ID,
  method="REML",
  digits=4,
  data=srES)
```

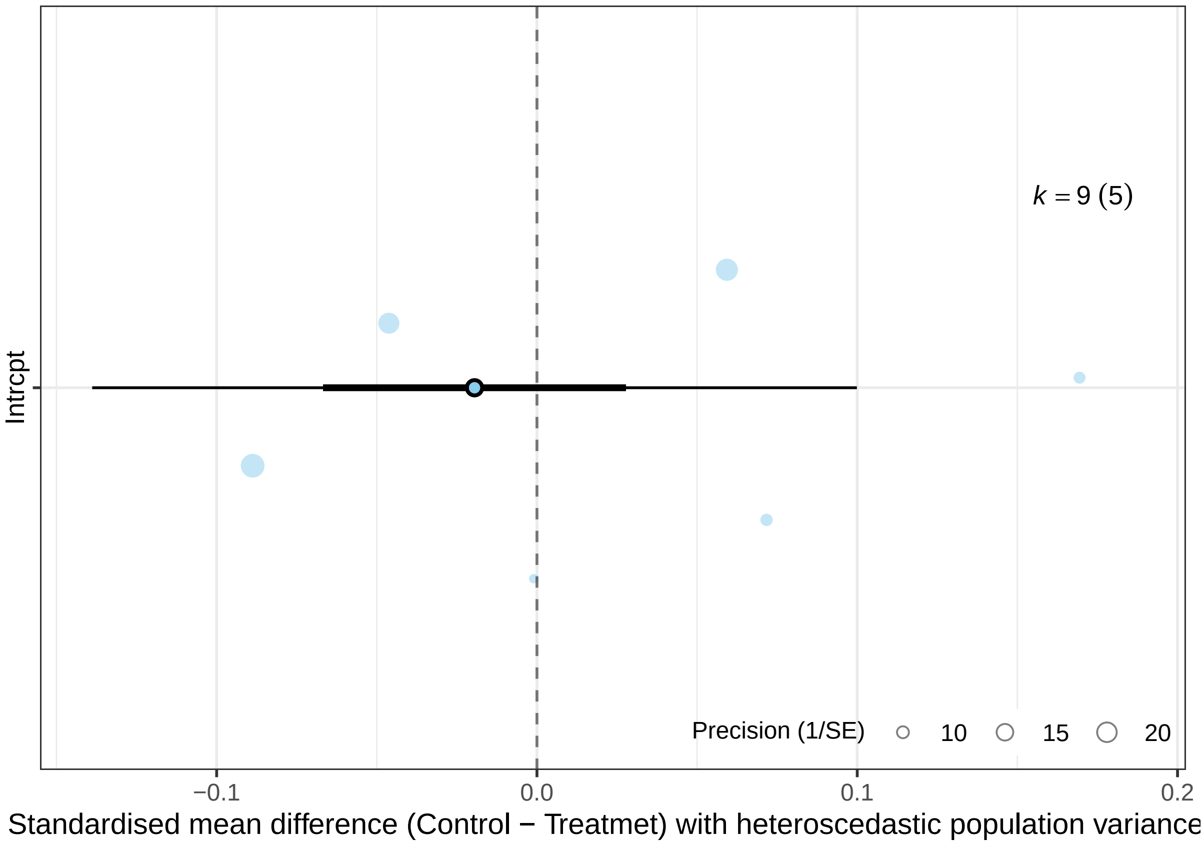
```
## Warning: There are outcomes with non-positive sampling variances.

## Warning: 'V' appears to be not positive definite.
```

```r
summary(sr1)
```

```
##
## Multivariate Meta-Analysis Model (k = 9; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##   8.8698  -17.7396  -11.7396  -11.5012   -5.7396
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed      factor
## sigma^2.1  0.0000  0.0000      5     no     PaperID
## sigma^2.2  0.0031  0.0559      9     no  PaperID/ID
##
## Model Results:
##
## estimate      se     zval    pval    ci.lb   ci.ub
##  -0.0195  0.0241  -0.8076  0.4193  -0.0668  0.0278
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# no significant difference
orchard_plot(sr1,
             group = "PaperID",
             xlab = "Standardised mean difference (Control - Treatmet) with heteroscedastic population 
```




```
# Try a one-variable with peatland type
# Although we don't really have enough data for this
sr2 <-rma.mv(
  yi=yi,
  V=vi,
  mods=~Peatland_type -1,
  random= ~1|PaperID/ID,
  method="REML",
  digits=4,
  data=srES)
```

```
## Warning: There are outcomes with non-positive sampling variances.

## Warning: 'V' appears to be not positive definite.
```

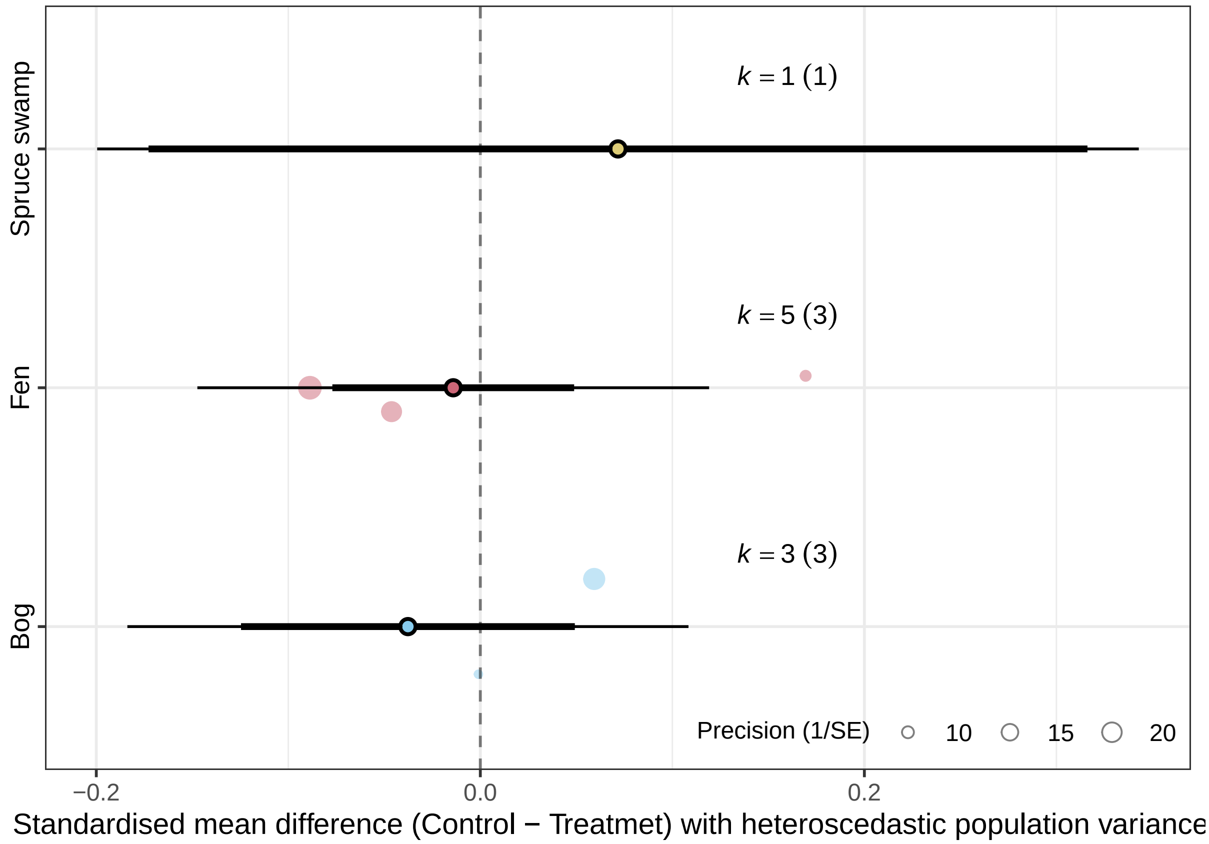
```
summary(sr2)
```

```
##
## Multivariate Meta-Analysis Model (k = 9; method: REML)
##
##    logLik   Deviance        AIC        BIC       AICc
##    6.3460   -12.6921    -2.6921    -3.7333    57.3079
##
## Variance Components:
```

```
##
##             estim    sqrt  nlvls  fixed      factor
## sigma^2.1  0.0000  0.0000      5     no     PaperID
## sigma^2.2  0.0036  0.0599      9     no  PaperID/ID
##
## Test of Moderators (coefficients 1:3):
## QM(df = 3) = 1.2460, p-val = 0.7420
##
## Model Results:
##
##                             estimate      se     zval    pval    ci.lb   ci.ub
## Peatland_typeBog             -0.0377  0.0443  -0.8502  0.3952  -0.1246  0.0492
## Peatland_typeFen             -0.0141  0.0321  -0.4392  0.6605  -0.0771  0.0489
## Peatland_typeSpruce swamp     0.0717  0.1247   0.5747  0.5655  -0.1728  0.3162
##
## Peatland_typeBog
## Peatland_typeFen
## Peatland_typeSpruce swamp
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# No significant effects
# bogs tended to decrease in SR with drawdown while other peatlands increased

orchard_plot(sr2,
             mod = "Peatland_type",
             group = "PaperID",
             xlab = "Standardised mean difference (Control - Treatmet) with heteroscedastic population v
```




```r
# Try continuous variable WTD difference
# note that one study didn't report WTD
# Try with WTD difference as another mod
sr3 <-rma.mv(
  yi=yi,
  V=vi,
  mods = ~ WTD_Difference,
  random= ~1|PaperID/ID,
  method="REML",
  digits=4,
  data=srES)
```

```
## Warning: There are outcomes with non-positive sampling variances.

## Warning: 'V' appears to be not positive definite.
```

```r
summary(sr3)
```

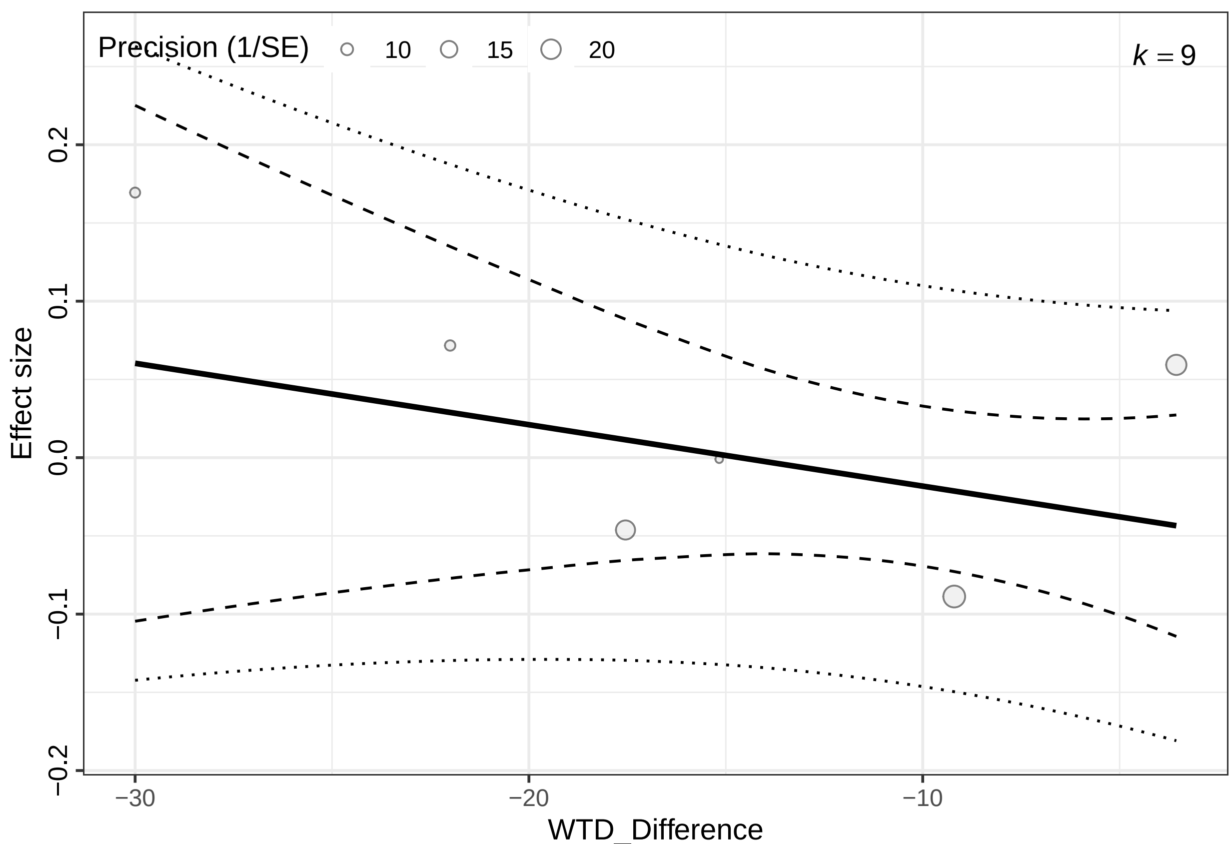
```
##
## Multivariate Meta-Analysis Model (k = 9; method: REML)
##
##    logLik   Deviance        AIC        BIC       AICc
##    7.9407   -15.8815    -7.8815    -8.0978    12.1185
##
```

```
## Variance Components:
##
##              estim    sqrt  nlvls  fixed      factor
## sigma^2.1  0.0000  0.0000      5     no     PaperID
## sigma^2.2  0.0036  0.0601      9     no  PaperID/ID
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 0.9698, p-val = 0.3247
##
## Model Results:
##
##                 estimate      se     zval    pval    ci.lb   ci.ub
## intrcpt          -0.0575  0.0473  -1.2151  0.2243  -0.1502  0.0352
## WTD_Difference   -0.0039  0.0040  -0.9848  0.3247  -0.0117  0.0039
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# no significant effect
# smaller differences in WTD had different

sr3_plot <- mod_results(sr3, mod = "WTD_Difference", group = "PaperID")
bubble_plot(sr3_plot, group = "PaperID",  mod = "WTD_Difference", xlab = "WTD_Difference")
```

```
## Ignoring unknown labels:
## * parse : "TRUE"
```



TO DO:

- Find outliers (https://doing-meta.guide/heterogeneity.html?q=case#outliers)
-